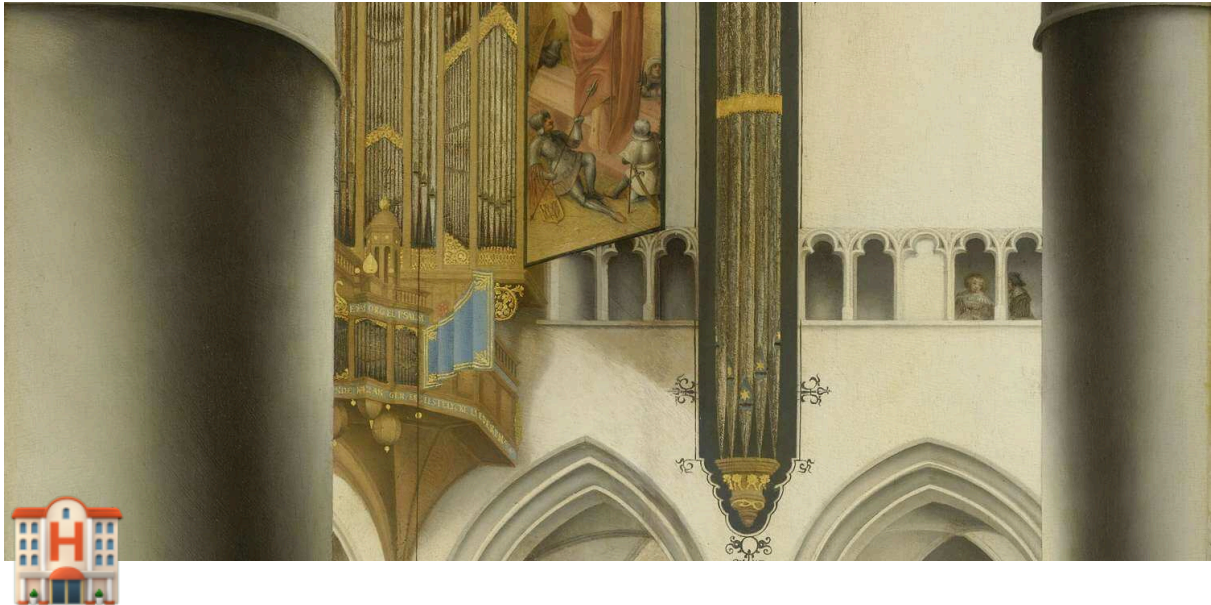




Hotel Management System

Introduction



The Hotel Management System is a comprehensive software solution designed to streamline and automate various aspects of hotel operations, including reservation management, room allocation, user authentication, and more. The system provides a robust backend infrastructure with a set of well-defined APIs to facilitate easy integration with front-end applications or external services. The project employs a relational database model to efficiently store and retrieve data, ensuring data consistency and integrity.

Objective



Hotel Management System aims to enhance project and task collaboration, improve productivity, and provide valuable insights into project progress and team performance. Its user-friendly interface and robust features make it a valuable tool for organizations seeking efficient task and project management.

Database Schema

▼ ER Diagram

[ER_Diagram_Hotel.mwb](#)

https://prod-files-secure.s3.us-west-2.amazonaws.com/31282dbc-a6a3-454c-930f-bedcf45363ec/db6df871-81d0-4827-a886-14d8542dd1d3/ER_Diagram_Hotel.pdf

▼ Database Script File

[create hotel.sql](#)

Design Guidelines

1. All Rest API requests and response formats must be in JSON.
2. 200, 201, 400, 404, 401, 500, 505 – use the appropriate HTTP status code in your response object.
3. Handle the Rest exceptions globally using the Rest exception handler for all endpoints.
4. Test your Rest Api using Junit and Mockito.

5. Participants should not alter database schema under any circumstances except wherever changes are mentioned explicitly.

Rest API Design

All REST API requests and responses follow the JSON format. The project implies standard HTTP status codes (200, 201, 400, 404, 401, 500, 505) appropriately in response objects to convey the outcome of API requests.

Status Codes

HTTP response codes are used to indicate general classes of success and error.

HTTP Status Quote	Description
200	Successfully processed request.
201	Created
202	Accepted
400	Bad Request
401	Unauthorized
403	Forbidden
404	Not Found
500	Internal Server Error

Endpoints

Endpoints

Name	Http Method	Description	Success Response	Error Response
/api/hotels/post	POST	Create a new Hotel	<pre>{ "code": "POSTSUCCESS", "message": "Hotel added successfully" }</pre>	<pre>{ "code": "ADDFAILS", "message": "Hotel already exist" }</pre>
/api/hotels/all	GET	Get a list of all hotels	{list of all hotels}	<pre>{ "code": "GETFAILS", "message": "Hotel list is empty" }</pre>
/api/hotels/{hotel_id}	GET	Get details of a specific hotel by ID.	{Object of hotel}	<pre>{ "code": "GETFAILS", "message": "Hotel doesn't exist" }</pre>
/api/hotels/{amenity_id}	GET	Get List of hotel which provide specific amenity	{List of hotel}	<pre>{ "code": "GETFAILS", "message": "No hotel is found with the specific amenity" }</pre>
/api/hotels/update/{hotel_id}	PUT	Update hotel details.	<pre>{ "code": "UPDATESUCCESS", "message": "Hotel updated successfully" }</pre>	<pre>{ "code": "UPDTFAILS", "message": "Hotel doesn't exist" }</pre>
/api/users/delete/{userId}	DELETE	Delete a user	<pre>{ "code": "DELETESUCCESS", "message": "User deleted successfully" }</pre>	<pre>{ "code": "DLTFAILS", "message": "User doesn't exist" }</pre>

Aa Name	⌘ Http Method	≡ Description	≡ Success Response	≡ Error Response
/api/RoomType/post	POST	Create a new roomType	<pre>{ "code": "POSTSUCCESS", "message": "roomType added successfully" }</pre>	<pre>{ "code": "ADDFAILS", "message":RoomT ype already exist" }</pre>
/api/RoomType/update/{RoomType_id}	PUT	Update RoomType details.	<pre>{ "code": "UPDATESUCCESS", "message": "RoomType updated successfully" }</pre>	<pre>{ "code": "UPDTFAILS", "message":RoomT ype doesn't exist exist" }</pre>
/api/projects/delete/{projectId}	DELETE	Delete a project.	<pre>{ "code": "DELETESUCCESS", "message": "Proejct deleted successfully" }</pre>	<pre>{ "code": "DLTFAILS", "message":Project doesn't exist exist" }</pre>
/api/RoomType/{RoomType_id}	GET	Get a Room Type by Id	{Get a RoomType Object}	<pre>{ "code": "GETFAILS", "message":RoomT ype doesn't exist " }</pre>
/api/rooms/post	POST	Create a new room .	<pre>{ "code": "POSTSUCCESS", "message": " room added successfully" }</pre>	<pre>{ "code": "ADDFAILS", "message":Room already exist" }</pre>
/api/room/all	GET	Get all rooms	{List of all rooms}	<pre>{ "code": "GETALLFAILS", "message":room }</pre>

Aa Name	⚙ Http Method	≡ Description	≡ Success Response	≡ Error Response
				<pre>list is empty" }</pre>
<u>/api/room/{room_id}</u>	GET	Get all a room	{get a room object}	<pre>{ "code": GETALLFAILS", "message":room doesn't exist" }</pre>
<u>/api/rooms/available/{roomTypeId}</u>	GET	Retrieve a list of available rooms of a specific type	{List of rooms}	<pre>{ "code": GETALLFAILS", "message": No room found with given type }</pre>
<u>/api/rooms/location/{location}</u>	GET	Retrieve rooms based on their location.	{List of rooms}	
<u>/api/rooms/amenities/{amenityId}</u>	GET	Retrieve rooms that have a specific amenity.	{List of rooms}	
<u>/api/rooms/{amenity_id}</u>	GET	Get all rooms which provides a specific amenity	{List of rooms}	<pre>{ "code": GETALLFAILS", "message": amenity does't exist with given id" }</pre>
<u>/api/room/update/{room_id}</u>	PUT	Update room details.	<pre>{ "code": "UPDATESUCCESS", "message": "Room updated successfully" }</pre>	<pre>{ "code": UPDTFAILS", "message":Room doesn't exist exist" }</pre>

Name	Http Method	Description	Success Response	Error Response
<u>/api/tasks/{task_id}</u>	DELETE	Delete a task.	<pre>{ "code": "DELETESUCCESS", "message": "task deleted successfully" }</pre>	<pre>{ "code": "DLTFAILS", "message":Task doesn't exist exist" }</pre>
Untitled				
<u>/api/reservation/post</u>	POST	Create a new reservation.	<pre>{ "code": "POSTSUCCESS", "message": "Reservation added successfully" }</pre>	<pre>{ "code": "ADDFAILS", "message":reserv ation already exist" }</pre>
<u>/api/reservation/all</u>	GET	Get a list of all reservation.	{List of reservations}	<pre>{ "code": "GETALLFAILS", "message":reserv ation list is empty" }</pre>
<u>/api/reservation/{reservation_id}</u>	GET	Get reservation By Id	{Object of reservation}	<pre>{ "code":GETFAILS", "message":Reserv ation doesn't exist" }</pre>
<u>/api/reservations/date-range?startDate={startDate}&endDate={endDate}</u>	GET	Retrieve reservations within a specified date range.	{List of Object of reservation}	
<u>/api/reservation/update/{reservation_id}</u>	PUT	Update reservation details.	<pre>{ "code": "UPDATESUCCESS", "message": "Reservation</pre>	<pre>{ "code": "UPDTFAILS", "message":Reserv ation doesn't</pre>

Aa Name	⌘ Http Method	≡ Description	≡ Success Response	≡ Error Response
			<code>updated successfully"</code>	<code>exist"</code>
<u>/api/reservation/{reservation_id}</u>	DELETE	Delete a reservation .	<code>{ "code": "DELETESUCCESS", "message": "reservation deleted successfully" }</code>	<code>{ "code": "DLTFAILS", "message": "Reservation doesn't exist exist" }</code>
Untitled				
<u>/api/amenity/post</u>	POST	Create a new amenity	<code>{ "code": "POSTSUCCESS", "message": "Amenity added successfully" }</code>	<code>{ "code": "ADDFAILS", "message": "Amenity already exist" }</code>
<u>/api/amenity/all</u>	GET	Get a list of all amenities .	{List of amenity }	<code>{ "code": "GETALLFAILS", "message": "Amenity list is empty" }</code>
<u>/api/amenity/{amenity_id}</u>	GET	Get amenity By Id	{Object of amenity}	<code>{ "code": "GETFAILS", "message": "Amenity doesn't exist" }</code>
<u>/api/amenity/{hotel_id}</u>	GET	Get all amenities which is provided by a specific hotel	{List of amenity }	<code>{ "code": "GETALLFAILS", "message": "hotel not found with given hotel id" }</code>
<u>/api/amenity/{room_id}</u>	GET	Get all amenities which is provided	{List of amenity }	<code>{ "code":</code>

Aa Name	⚙ Http Method	≡ Description	≡ Success Response	≡ Error Response
		by a room		GETALLFAILS", "message":room not found with given hotel id" }
/api/amenity/{amenity_id}	DELETE	Delete a amenity.	{ "code": "DELETESUCCESS", "message": "Amenity deleted successfully" }	{ "code": DLTFAILS", "message":Amenity doesn't exist exist" }
/api/amenity/update/{amenity_id}	PUT	Update amenity details.	{ "code": "UPDATESUCCESS", "message": "Amenity updated successfully" }	{ "code": UPDTFAILS", "message":Amenity doesn't exist" }
/api/payment/post	POST	Create a new payment	{ "code": "POSTSUCCESS", "message": "Payment added successfully" }	{ "code": ADDFAILS", "message":Payme nt already exist" }
/api/payment/all	GET	Get a list of all payments.	{List of payment}	{ "code": GETALLFAILS", "message":Payme nt list is empty" }
/api/payment/{payment_id}	GET	Get payment By Id	{Object of payment}	{ "code":GETFAILS", "message":Payme nt doesn't exist" }

Name	Http Method	Description	Success Response	Error Response
/api/payments/status/{status}	GET	Retrieve payments based on their status (e.g., "Paid," "Pending").	{List of payment}	{ "code": GETALLFAILS", " "message":Payment list is empty" }
/api/payments/total-revenue	GET	Retrieve the total revenue generated through payments.	{Amount of revenue}	
/api/payment/{payment_id}	DELETE	Delete a payment.	{ "code": "DELETESUCCESS", " "message": "Payment deleted successfully" }	{ "code": DLTFAILS", " "message":Payment doesn't exist exist" }
/api/review/post	POST	Upload a new review.	{ "code": "POSTSUCCESS", " "message": "Review added successfully" }	{ "code": ADDFAILS", " "message":Review already exist" }
/api/review/all	GET	Get a list of all reviews.	{List of object of Review}	{ "code": GETALLFAILS", " "message":Review list is empty" }
/api/review/{review_id}	GET	Get details of a specific review by ID.	{Object of Review}	{ "code": GETFAILS", " "message":Review doesn't exist" }
/api/reviews/rating/{rating}	GET	Retrieve reviews with a specific rating.	{List of Reviews}	{ "code": GETALLFAILS", "

Name	Http Method	Description	Success Response	Error Response
				<pre>"message":Review list is empty" }</pre>
/api/reviews/recent	GET	Retrieve the most recent reviews.	{Object of Reviews}	<pre>{ "code": GETFAILS", "message":Review doesn't exist" }</pre>
/api/review/update/{review_id}	PUT	Update review details.	<pre>{ "code": "UPDATESUCCESS", "message": "Review updated successfully" }</pre>	<pre>{ "code": UPDTFAILS", "message":Review doesn't exist exist" }</pre>
/api/review/delete/{review_id}	DELETE	Delete an review.	<pre>{ "code": "DELETESUCCESS", "message": "Review deleted successfully" }</pre>	<pre>{ "code": DLTFAILS", "message":Review doesn't exist" }</pre>
Untitled				
/api/hotelamenity/post	POST	Create a new hotelamenity.	<pre>{ "code": "POSTSUCCESS", "message": "Hotelamenity added successfully" }</pre>	<pre>{ "code": ADDFAILS", "message":Hotela menity already exist" }</pre>
/api/roomAmenity/post	POST	Create a new roomamenity	<pre>{ "code": "POSTSUCCESS", "message": "RoomAmenity added successfully" }</pre>	<pre>{ "code": ADDFAILS", "message":Room Amenity already exist" }</pre>

